Then, we get $E_4\left(E_3\left(E_2\left(E_1A\right)\right)\right)=\begin{bmatrix}1 & \frac{1}{2}\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I_2$.

We write the above sequence of elementary transformations in the following manner:

$$A=\begin{bmatrix}2 & -1\\ 3 & 4\end{bmatrix}\xrightarrow{R_1\to\left(\frac{1}{2}\right)R_1}\begin{bmatrix}1 & \frac{-1}{2}\\ 3 & 4\end{bmatrix}\xrightarrow{R_2\to R_2+(-3)R_1}\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & \frac{11}{2}\end{bmatrix}\xrightarrow{R_2\to\left(\frac{2}{11}\right)R_2}\begin{bmatrix}1 & -\frac{1}{2}\\ 0 & 1\end{bmatrix}\xrightarrow{R_1\to R_1+\left(\frac{1}{2}\right)R_2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Example 1.19

Show that the matrix $\begin{bmatrix}3 & 1 & 4\\ 2 & 0 & -1\\ 5 & 2 & 1\end{bmatrix}$ is non-singular and reduce it to the identity matrix by elementary row transformations.

Solution

Let $A=\begin{bmatrix}3 & 1 & 4\\ 2 & 0 & -1\\ 5 & 2 & 1\end{bmatrix}$. Then, $|A|=3(0+2)-1(2+5)+4(4-0)=6-7+16=15\neq 0$. So, A is non-singular. Keeping the identity matrix as our goal, we perform the row operations sequentially on A as follows:

$$\begin{bmatrix}3 & 1 & 4\\ 2 & 0 & -1\\ 5 & 2 & 1\end{bmatrix}\xrightarrow{R_1\to\frac{1}{3}R_1}\begin{bmatrix}1 & \frac{1}{3} & \frac{4}{3}\\ 2 & 0 & -1\\ 5 & 2 & 1\end{bmatrix}\xrightarrow{R_2\to R_2-2R_1,\ R_3\to R_3-5R_1}\begin{bmatrix}1 & \frac{1}{3} & \frac{4}{3}\\ 0 & -\frac{2}{3} & -\frac{11}{3}\\ 0 & \frac{1}{3} & -\frac{17}{3}\end{bmatrix}\xrightarrow{R_2\to\left(-\frac{3}{2}\right)R_2}\begin{bmatrix}1 & \frac{1}{3} & \frac{4}{3}\\ 0 & 1 & \frac{11}{2}\\ 0 & \frac{1}{3} & -\frac{17}{3}\end{bmatrix}$$

$$\xrightarrow{R_1\to R_1-\frac{1}{3}R_2,\ R_3\to R_3-\frac{1}{3}R_2}\begin{bmatrix}1 & 0 & -\frac{1}{2}\\ 0 & 1 & \frac{11}{2}\\ 0 & 0 & -\frac{15}{2}\end{bmatrix}\xrightarrow{R_3\to\left(-\frac{2}{15}\right)R_3}\begin{bmatrix}1 & 0 & -\frac{1}{2}\\ 0 & 1 & \frac{11}{2}\\ 0 & 0 & 1\end{bmatrix}\xrightarrow{R_1\to R_1+\frac{1}{2}R_3,R_2\to R_2-\frac{11}{2}R_3}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}.$$

1.3.4 Gauss-Jordan Method

Let A be a non-singular square matrix of order n. Let B be the inverse of A.

Then, we have $AB=BA=I_n$. By the property of I_n, we have $A=I_nA=AI_n$.

Consider the equation $A=I_nA$ …(1)

Since A is non-singular, pre-multiplying by a sequence of elementary matrices (row operations) on both sides of (1), A on the left-hand-side of (1) is transformed to the identity matrix I_n and the same sequence of elementary matrices (row operations) transforms I_n of the right-hand-side of (1) to a matrix B. So, equation (1) transforms to $I_n=BA$. Hence, the inverse of A is B. That is, $A^{-1}=B$.

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